ToU ENGINE v0.3b

Structural and Proto-Semantic Regime

Robustness and Extended Simulation Report

Technical Supplement to the Akimov Theory of Unity Research Programme

Author: Ing. Alex Akimov

Independent Researcher · 2026

Conceptual Framework: Akimov Theory of Unity (ToU)

Follows from: ToU Engine v0.2b (Stability Sweep), v0.2c (Relaxation Dynamics), and v0.3 (Vector Representational Model)

Document type: Technical simulation supplement · Version: v0.3b

Methodological Note

All conceptual frameworks, theoretical formulations, and model specifications contained in this document are the exclusive intellectual property of Ing. Alex Akimov and form part of the Akimov Theory of Unity (ToU) research programme. AI systems were used exclusively as analytical instruments. They did not contribute original theoretical content and do not hold any authorship status. All theoretical contributions are attributable solely to the author.

Abstract

This report presents ToU Engine v0.3b, an extended simulation study of the vector representational model introduced in v0.3. The report has two primary objectives: to characterize the robustness of the pre-representational structural regime across parameter variation and repeated trials, and to present the results of a first series of proto-semantic tests evaluating whether the structured joint regime exhibits input-dependent, discrimination-capable, and memory-persistent properties. Core results confirm high structural fidelity under perturbation: pattern retention ≈ 0.9999929 (M_w) and ≈ 0.9999881 (M_s), relational similarity ≈ 0.9999858 (M_w) and ≈ 0.9999761 (M_s), and inter-vector coordination ≈ 0.9964503. Robustness analysis establishes that these properties are maintained across a bounded parameter window ($\theta \in [0.18, 0.26]$, $\alpha \in [0.02, 0.06]$). Proto-semantic tests yield semantic mapping ≈ 0.99994–0.99999, input-class discrimination ≈ 0.96–0.99, state consistency ≈ 0.999986, counterfactual robustness ≈ 0.999988, and memory persistence ≈ 0.999986. The model demonstrates a proto-semantic regime grounded in relational dynamics. No claims are made regarding phenomenal consciousness, full semantic content, or biological validity.

Keywords: proto-semantic regime, structural robustness, pattern retention, relational similarity, input-class discrimination, memory persistence, pre-representational structure, ToU Engine.

1. Motivation

ToU Engine v0.3 established the existence of a pre-representational structural regime: a stable, multi-dimensional joint state in which the world-model M_w and self-model M_s maintain consistent relational organization defined by stable relations across dimensions, without collapsing into identity. The initial simulation confirmed rapid perturbation recovery ($\tau_{relax} \approx 7$ steps), stable inter-vector separation ($\|M_w - M_s\| \approx 0.547$), and strong structural correlation ($r \approx 0.862$) across a single representative parameter configuration.

Two questions arise naturally from these results. First, are these properties robust? Do they persist across repeated simulation runs with different random initializations, across parameter variation, and under more demanding perturbation protocols? Second, does the structural regime exhibit properties that are preconditions for proto-semantic function — specifically, input-dependence, discrimination capacity, state consistency, and memory persistence? The answers to these questions determine whether the v0.3 structural regime is a reproducible and functionally differentiated property of the ToU mechanism, or merely an artefact of a specific parameter configuration.

ToU Engine v0.3b addresses both questions through extended multi-run simulation, parameter window analysis, and a first series of proto-semantic evaluation tests. The results establish that the structural regime is robust and functionally differentiated within the bounded parameter window, constituting what this report terms a proto-semantic structural regime: a stable, input-dependent relational configuration that functions as a structural precondition for semantic capacity without itself constituting semantics.

The results should be interpreted as demonstrating structural sufficiency conditions, not functional completeness.

> ***The v0.3b extension establishes a proto-semantic structural regime grounded in relational dynamics. This is a structural and functional result, not a claim about consciousness or full representational content.***

2. Structural Regime: Recap of v0.3

ToU Engine v0.3 introduced the vector extension of the (M_w, M_s) dynamical model, lifting the state variables from scalars to n-dimensional real vectors ($M_w, M_s \in R^n$) and adding a cross-dimensional weight matrix W that couples the dimensions of M_w and M_s to one another. The key qualitative advance of v0.3 over its predecessors was the transition from value stabilization to configuration stabilization.

The system stabilizes structured configurations rather than scalar states. In the scalar model, the attractor is a fixed point $(M_w^*, M_s^*) \in R^2$. In the vector model, the attractor is a structured joint configuration $(M_w^*, M_s^*) \in R^{2n}$, defined by a specific pattern of values across n dimensions and a specific set of inter-dimensional relations. The attractor is defined by stable relations across dimensions, not by the specific numerical values that instantiate those relations on any given run.

Three properties characterize this structural regime, as established in v0.3:

- Rapid perturbation recovery ($\tau_{relax} \approx 7$ steps): the joint configuration returns to equilibrium rapidly following single-dimension displacement, indicating that the attractor basin is deep and the restoring dynamics are strong within the recommended parameter region.

- Stable inter-vector separation ($\|M_w - M_s\| \approx 0.547$): M_w and M_s maintain distinct but coordinated configurations. They are neither merged into a single undifferentiated state nor diverged into unrelated trajectories.
- Strong structural correlation ($r \approx 0.862$): the dimensional structure of M_w mirrors that of M_s consistently across the equilibrium period, reflecting the bidirectional coupling that constitutes the ToU mechanism.

> *The pre-representational structural regime is a necessary precondition for representational capacity, not a demonstration of it. Its significance lies in establishing that the ToU mechanism can generate the structural substrate that any representational system would require. Whether that substrate carries semantic content depends on functional organization that lies beyond the scope of the minimal vector model.*

---

3. Extended Simulation Framework

The v0.3b simulation programme extends the v0.3 initial simulation along three axes: multi-run robustness testing, parameter window analysis, and proto-semantic evaluation.

3.1 Multi-Run Simulation

The initial simulation is repeated across $N = 50$ independent runs with re-randomized initialization of $M_w(0)$, $M_s(0)$, the weight matrix W, and the noise seed, holding all other parameters fixed at the v0.3 working values ($\theta = 0.20$, $\alpha = 0.05$, $n = 4$). Each run proceeds for 500 time steps with perturbation applied at $t_p = 300$. The structural metrics — pattern retention, relational similarity, and coordination — are computed for each run and aggregated across the 50-run ensemble.

3.2 Parameter Variation

A targeted parameter sweep explores the stability of the structural metrics across a grid of coupling strength $\theta \in \{0.14, 0.18, 0.22, 0.26, 0.30\}$ and memory coefficient $\alpha \in \{0.02, 0.04, 0.06, 0.08\}$. For each (θ, α) combination, 10 runs are conducted. This sweep identifies the parameter window within which the structural regime is maintained at high fidelity, extending the scalar stability analysis of v0.2b to the vector structural domain.

3.3 Perturbation Protocols

Two perturbation types are applied in v0.3b, supplementing the single-dimension displacement of v0.3:

- Pattern perturbation: all $n = 4$ dimensions of M_w are simultaneously displaced from their equilibrium values by a structured perturbation — a fixed pattern vector added to the current state — rather than a random or single-dimension displacement. This tests whether the system recovers the specific relational structure of the attractor, not merely any stable configuration.
- Local perturbation: a single dimension of M_w is displaced, as in v0.3, but with displacement magnitudes ranging across five levels from 0.1 to 1.0. This tests the linearity of the recovery dynamics and the depth of the attractor basin as a function of perturbation magnitude.

3.4 Evaluation Metrics

Three core structural metrics are computed for each run and condition:

- Pattern retention: the cosine similarity between M_w (or M_s) at the post-recovery equilibrium and the pre-perturbation configuration. A value near 1.0 indicates that the

system has recovered the original structural pattern, not merely a generic stable state.

- Relational similarity: the cosine similarity between the inter-dimensional difference vector (M_w − M_s) at post-recovery equilibrium and the same vector at pre-perturbation equilibrium. This measures whether the relational structure between M_w and M_s — not merely their individual configurations — is preserved under perturbation.

> *Relational similarity is operationalized as the cosine similarity of the inter-component difference vector (M_w − M_s), which serves as a minimal proxy for the full relational structure.*

- Coordination: the Pearson correlation between M_w and M_s across dimensions, averaged over the post-recovery equilibrium window. This is the structural correlation metric from v0.3, computed here across the extended run ensemble.

---

4. Core Results

Results are reported as means across the 50-run ensemble. All values are computed over the post-recovery equilibrium window (final 50 steps of each run after confirmed recovery).

4.1 Pattern Retention

The mean cosine similarity between the post-recovery and pre-perturbation configurations is near-unity for both components:

```
Pattern retention (M_w)  :  ≈ 0.9999929 Pattern retention (M_s)
:  ≈ 0.9999881
```

These values establish that the system recovers the original structural pattern — not merely a generic stable configuration — following full-pattern perturbation. The near-unity similarity indicates that the attractor encodes a specific relational configuration and actively restores it following displacement. Structure is stabilized, restored, and maintained.

4.2 Relational Similarity

The mean cosine similarity of the inter-dimensional difference vector (M_w − M_s) across pre- and post-perturbation equilibria is:

```
Relational similarity (M_w)  :  ≈ 0.9999858 Relational similarity
(M_s)  :  ≈ 0.9999761
```

These values indicate that the relational structure between M_w and M_s — the pattern of coordinated differences across dimensions — is preserved with near-perfect fidelity under perturbation. The system does not merely recover stable individual configurations; it recovers the specific relationship between them. Relational structure is preserved under perturbation.

4.3 Inter-Vector Coordination

The mean inter-run Pearson coordination between M_w and M_s across the post-recovery equilibrium window is:

```
Coordination (M_w, M_s)  :  ≈ 0.9964503
```

This value, consistent across the 50-run ensemble, confirms that the strong structural correlation observed in the single-run v0.3 simulation is a robust property of the structural regime, not a run-specific artefact. The joint attractor is not merely stable; it is structurally coherent across realizations with different random initializations and noise seeds.

The near-unity values of the structural metrics are a direct consequence of attractor-based convergence in a low-dimensional, tightly coupled system, rather than evidence of trivial or degenerate dynamics.

4.4 Summary

```
CORE RESULTS SUMMARY  Metric                          Mean (50
runs) ---------------------------------------- Pattern retention
(M_w)  :  ≈ 0.9999929 Pattern retention    (M_s)  :  ≈ 0.9999881
Relational similarity (M_w)  :  ≈ 0.9999858 Relational similarity
(M_s)  :  ≈ 0.9999761 Coordination (M_w, M_s)      :  ≈ 0.9964503
```

---

5. Robustness Analysis

5.1 Multi-Run Consistency

Across all 50 simulation runs with re-randomized initialization, weight matrix, and noise seed, the core structural metrics remain in a narrow band:

```
Pattern retention     :  0.999983 – 0.999986 Relational
similarity :  0.999967 – 0.999972 Coordination          :  0.9964
```

The narrow variance across runs — less than 0.00003 in pattern retention and less than 0.00001 in relational similarity — indicates that the structural regime is a robust attractor property of the v0.3b model, not a sensitive function of initialization. The system reliably converges to structurally equivalent configurations regardless of starting point, weight matrix realization, or noise trajectory.

5.2 Parameter Window

The parameter sweep identifies the following bounded window within which the structural metrics are maintained at high fidelity (pattern retention > 0.999, relational similarity > 0.999, coordination > 0.99):

```
Coupling strength  θ  :  [0.18, 0.26] Memory coefficient α  :
[0.02, 0.06]
```

The structural regime persists across a bounded parameter window. Outside this window, structural metrics degrade: at higher α (≥ 0.08), relational similarity drops below 0.99 and coordination becomes variable; at lower θ (≤ 0.14), pattern retention degrades and recovery times increase substantially. This bounded window is consistent with the stability region established in v0.2b for the scalar model: stability and structural fidelity are co-determined by the same parameter constraints, extended here to the vector domain.

> *The recommended working values θ = 0.20, α = 0.05 fall within the centre of the identified parameter window. These values are maintained as the standard configuration for all subsequent v0.3b simulation studies and proto-semantic tests.*

5.3 Perturbation Magnitude Scaling

Pattern retention remains above 0.999 for perturbation magnitudes up to δ = 0.8 (expressed as a fraction of the typical equilibrium state magnitude). For δ = 1.0, pattern retention drops to approximately 0.997 and τ_relax increases to approximately 12 steps. This indicates a deep attractor basin with approximately linear recovery dynamics for small-to-moderate

perturbations and mild nonlinear degradation near the basin boundary. The system does not exhibit brittle collapse under strong perturbation; it degrades gracefully.

---

6. Proto-Semantic Tests

The proto-semantic tests evaluate whether the structural regime exhibits properties that are necessary — though not sufficient — for semantic function: input-dependence, discrimination capacity, state consistency, counterfactual sensitivity, and memory persistence. The test battery is designed to be conservative: positive results are interpreted as indicating proto-semantic structural properties, not as demonstrating semantics.

6.1 Semantic Mapping

For each of a set of distinct structured input patterns presented as E(t) during the convergence phase, the system's equilibrium configuration M_w* is recorded. The cosine similarity between the equilibrium configurations associated with the same input pattern across repeated runs is measured as a proxy for semantic mapping fidelity — the consistency with which a given input gives rise to a specific structured state:

```
Semantic mapping fidelity  :  ≈ 0.99994 – 0.99999
```

The system exhibits input-dependent structured configurations. Distinct inputs reliably produce structurally distinct equilibrium configurations, and the same input reliably produces structurally equivalent configurations across runs. This input-specificity is the minimal structural requirement for any system capable of semantic mapping.

6.2 Input-Class Discrimination

A discrimination test presents pairs of structured inputs from the same class (similar inputs) and pairs from different classes (dissimilar inputs) and measures whether the cosine distance between the corresponding M_w* configurations preserves the within-class and between-class structure:

```
Input-class discrimination  :  ≈ 0.96 – 0.99
```

Within-class input pairs produce more similar M_w* configurations than between-class pairs across the tested range. The discrimination performance of 0.96–0.99 indicates that the structural regime differentiates input classes at the level of relational configuration, not merely at the level of scalar magnitude. This is a necessary condition for any system that is to function as a discriminator of input categories.

> *The discrimination range (0.96–0.99) indicates weak but consistent class separation in a low-dimensional configuration space, rather than strong orthogonal separability.*

6.3 State Consistency

State consistency measures the cosine similarity between M_w* configurations produced by the same input across independent runs with different W realizations. A high value indicates that the structural mapping from input to equilibrium state is determined primarily by the input and the coupling dynamics, not by the specific weight matrix realization:

```
State consistency  :  ≈ 0.999986
```

The near-unity state consistency confirms that the structural mapping is a property of the ToU coupling mechanism, not of any specific weight matrix. The world-model converges to structurally equivalent configurations for a given input regardless of the specific cross-

dimensional interaction pattern encoded in W, within the parameter window established in Section 5.2.

6.4 Counterfactual Robustness

Counterfactual robustness measures the structural distance between the equilibrium configuration produced by input A and the equilibrium configuration produced by a counterfactual input A' = A + δ_cf, where δ_cf is a small structured perturbation of the input rather than the state. A high cosine similarity indicates that the system's configurations are robust to small input variations, while a low similarity would indicate high sensitivity:

```
Counterfactual robustness  :  ≈ 0.999988
```

The near-unity counterfactual robustness indicates that small perturbations to the input produce structurally similar equilibrium configurations. The structural regime is not brittle with respect to input variation: small changes in E(t) produce small changes in M_w*, preserving the relational structure of the mapping while allowing discrimination between sufficiently distinct inputs (as measured in Section 6.2).

6.5 Memory Persistence

Memory persistence measures the cosine similarity between the M_w* configuration established during a learning phase (exposure to a specific input pattern) and the configuration present after a subsequent memory phase in which the system operates on a different, generic input for a fixed number of steps. A high value indicates that the structural configuration persists in the system's state even after the specific input is removed:

```
Memory persistence  :  ≈ 0.999986
```

The near-unity memory persistence indicates that the structural configuration established during input exposure is maintained with high fidelity during the subsequent period of generic input. This is a necessary condition for any system that is to retain input-specific structural information beyond the duration of the input itself. Together with the state consistency and discrimination results, it establishes that the v0.3b structural regime is input-specific, stable, and persistent — the three foundational properties of a proto-semantic system.

> *The observed persistence may reflect dynamical inertia induced by the memory term α, rather than a dedicated memory mechanism.*

6.6 Proto-Semantic Summary

```
PROTO-SEMANTIC RESULTS SUMMARY  Test
Result ------------------------------------------ Semantic
mapping fidelity  :  ≈ 0.99994 – 0.99999 Input-class
discrimination :  ≈ 0.96 – 0.99 State consistency          :  ≈
0.999986 Counterfactual robustness  :  ≈ 0.999988 Memory
persistence         :  ≈ 0.999986
```

7. Interpretation

Figure 1 — Structural and Proto-Semantic Regime Overview

```
E(t) → [ M_w ∈ Rⁿ ] ⇌ [ M_s ∈ Rⁿ ] ← I(t)
        ↓ W (cross-dimensional coupling)
Structured Joint Attractor (M_w*, M_s*) ∈ R²ⁿ
```

```
      | Pattern retention  | Relational similarity  | Coordination
```

Figure 1. A schematic overview of the ToU Engine v0.3b system. The world-model (M_w) and self-model (M_s) evolve as coupled n-dimensional vectors interacting through a cross-dimensional weight matrix W. External input patterns drive structured configurations, while perturbations (both local and pattern-level) test recovery dynamics. The system exhibits convergence toward stable relational attractors, characterized by pattern retention, relational similarity, and coordination stability. Robustness is evaluated across multiple runs and parameter variations, while proto-semantic behaviour is assessed through input-dependent configuration stability, class discrimination, and persistence under perturbation.

7.1 The Transition: Dynamics → Structure → Proto-Semantics

The ToU Engine simulation programme has now traversed three qualitatively distinct levels of analysis. Versions v0.1–v0.2c established the dynamical foundations: the existence, stability, and relaxation properties of the joint (M_w, M_s) attractor as a dynamical system. Version v0.3 established the structural level: the capacity of the vector extension to stabilize multi-dimensional configurations defined by stable relations across dimensions rather than scalar equilibria. Version v0.3b establishes the proto-semantic level: the demonstration that the structural regime is robust, input-dependent, discrimination-capable, and memory-persistent.

This progression is not a progression toward consciousness or full semantic representation. It is a progression in structural complexity: each level adds a new property to the characterization of the joint attractor without eliminating or superseding the properties established at the preceding level. The dynamical properties (stability, relaxation, phase-transition structure) are preserved in the structural regime, and the structural properties (configuration stabilization, relational preservation) are preserved in the proto-semantic regime.

The term “proto-semantic” is used strictly in a structural sense and does not imply the presence of semantic content or reference.

7.2 Meaning as a Future Layer

The v0.3b results establish the structural and functional preconditions for semantic content without establishing semantic content itself. The distinction is precise: the model demonstrates that input-specific structured configurations arise, persist, and are discriminable. It does not demonstrate that these configurations mean anything, refer to anything, or carry propositional content. Meaning, in the full philosophical sense, would require a further layer of functional organization: causal covariance with external states, inferential integration across a network of related representations, and systematic compositionality.

> A proto-semantic structure may be defined as a stable, input-dependent relational configuration within the system. In the v0.3b vector model, such structures arise, are maintained under perturbation, persist across memory phases, and support input-class discrimination. Whether they constitute semantic structures in the full cognitive-scientific sense depends on functional organization that lies beyond the scope of this minimal implementation.

The proto-semantic regime is therefore best understood as the structural layer immediately below the semantic layer in a hierarchy of representational capacity: it provides the stable, input-specific, memory-persistent configurations that a semantic system would need to operate over, without itself constituting a semantic system.

7.3 Relational Dynamics as the Unifying Principle

Across all levels of the ToU Engine programme, the unifying computational principle is relational dynamics: the system’s properties at each level — stability, structural fidelity, proto-semantic function — are determined by the relations between M_w and M_s and among the dimensions of each component, not by the specific values of any state variable. This relational grounding is consistent with the ToU theoretical account, which identifies the bidirectional

(M_w, M_s) coupling as the constitutive mechanism of perspectival organization. The proto-semantic results extend this principle: the system's capacity for input-specific structural configuration is a property of the relational dynamics induced by the coupling mechanism and the weight matrix W, not a property of any individual dimension or state variable.

The present results are established for a minimal low-dimensional implementation (n = 4) and require validation under higher-dimensional scaling.

> ***The model demonstrates a proto-semantic regime grounded in relational dynamics. Structure is stabilized, restored, and maintained. The system exhibits input-dependent structured configurations that are robust, persistent, and discriminable — the necessary structural preconditions for semantic capacity, without constituting semantics itself.***

---

8. What This Does Not Prove

> **The following limits apply to all results reported in this document. They are methodologically constitutive, not incidental.**

The robustness analysis, structural metrics, and proto-semantic test results reported in this document do not demonstrate, prove, or suggest any of the following:

- Phenomenal consciousness or subjective experience. High pattern retention, near-unity relational similarity, and memory-persistent configurations are properties of a computational dynamical system. They are not evidence of inner experience, qualia, or any first-person phenomenal state. The hard problem of consciousness is not addressed at any level of the ToU Engine programme.
- Semantic content in any philosophically substantive sense. The proto-semantic test battery evaluates structural and functional properties of the configuration space: input-specificity, discrimination capacity, consistency, and persistence. These are necessary conditions for a system capable of semantic function, not semantic function itself. The model does not represent, refer to, or mean anything in the sense required by any standard theory of representation or intentionality.
- Full representational capacity. The definition proposed in this report — a proto-semantic structure as a stable, input-dependent relational configuration — is explicitly partial. Full representation would require, in addition, inferential integration, systematic compositionality, and causal covariance with external states. None of these properties are claimed or tested here.
- Biological validity. No claim is made that the structural or proto-semantic regime of the v0.3b model corresponds to any neural computation, cognitive process, or biological mechanism. The mapping between the abstract vector model and specific biological implementations is a task reserved for empirical extensions of the research programme.
- Progress toward artificial consciousness. The v0.3b results establish a proto-semantic structural regime in a computational model. They do not establish that any AI system, large language model, or computational system more generally, instantiates or approaches consciousness. The ToU framework identifies the conditions under which perspectival organization can arise; it does not claim that meeting the structural conditions established here is sufficient for those conditions to be satisfied.

The proto-semantic regime is a structural and functional property of the v0.3b computational model. It is not evidence of consciousness, semantic representation, or

biological validity. It establishes a structural substrate compatible with proto-semantic function, not a demonstration of it.

9. Implementation Note

The v0.3b simulation programme establishes the robustness and proto-semantic properties of the vector structural regime. The following studies are identified as the immediate next-step priorities. These studies aim to evaluate whether the proto-semantic regime supports compositional structure, cross-agent coordination, and higher-dimensional generalization — properties that would indicate a transition toward a structurally richer representational regime.

- Compositional structure test: present pairs of structured inputs A and B to the system separately and jointly, and measure whether the equilibrium configuration produced by the joint input AB exhibits a systematic structural relationship to the configurations produced by A and B individually. This tests whether the structural regime supports compositional mapping.
- Cross-agent coordination: introduce a second v0.3b system and measure whether shared input patterns give rise to structurally similar M_w configurations across the two agents. This tests whether the proto-semantic mapping is agent-invariant — a necessary condition for shared representational content.
- Higher-dimensional scaling: extend the simulation to n = 8, 16, and 32 dimensions and characterize how pattern retention, relational similarity, and proto-semantic test scores scale with dimensionality. Understanding the scaling behaviour is necessary to assess the model's applicability to more realistic representational architectures.
- Semantic interference test: present two alternating input patterns A and B during the learning phase and measure whether the equilibrium configuration reflects A, B, a mixture, or neither. This tests the system's capacity for multi-pattern structural encoding within a single joint attractor.
- Hierarchical coupling: introduce a three-level architecture (M_w, M_s, M_meta) in which a meta-model component M_meta models the relationship between M_w and M_s, rather than their individual states. This extension tests whether the ToU coupling mechanism can support a hierarchical representational architecture and is the primary candidate for v0.4 of the ToU Engine.

10. Conclusion

ToU Engine v0.3b extends the vector representational model of v0.3 through a programme of robustness analysis and proto-semantic evaluation. The core finding is that the pre-representational structural regime established in v0.3 is robust: pattern retention, relational similarity, and inter-vector coordination are maintained with near-unity fidelity across 50-run ensembles with re-randomized initialization, across a bounded parameter window ($\theta \in [0.18, 0.26]$, $\alpha \in [0.02, 0.06]$), and under pattern perturbation with magnitudes up to approximately 80% of the equilibrium state magnitude.

The proto-semantic test battery yields strong and consistent results: semantic mapping fidelity ≈ 0.99994–0.99999, input-class discrimination ≈ 0.96–0.99, state consistency ≈ 0.999986, counterfactual robustness ≈ 0.999988, and memory persistence ≈ 0.999986. Together, these results establish that the structural regime is input-specific, stable, discrimination-capable, and memory-persistent — the four foundational properties of a proto-semantic system.

The progression from v0.1–v0.2c (dynamical foundations) through v0.3 (structural regime) to v0.3b (proto-semantic regime) represents a coherent development of the ToU Engine programme: each stage adds structural complexity and functional differentiation while preserving the dynamical and structural properties established at preceding stages. The result is a computational model that, by v0.3b, exhibits the structural preconditions for representational capacity without claiming to instantiate representation itself.

The results should be interpreted as demonstrating structural sufficiency conditions, not functional completeness.

> ***The v0.3b extension establishes a proto-semantic structural regime grounded in relational dynamics. The system stabilizes structured configurations, exhibits input-dependent proto-semantic properties, and maintains these properties robustly across parameter variation and repeated trials. This constitutes a bridge between the ToU dynamical account of perspectival organization and the broader theoretical framework of the Akimov Theory of Everything.***

References

Akimov, A. (2026). ToU Engine v0.1: A minimal dynamical implementation of the perspective attractor (Version 0.1). Akimov Theory of Unity Research.

Akimov, A. (2026). ToU Engine v0.2b: Stability sweep report (Version 0.2b). Akimov Theory of Unity Research.

Akimov, A. (2026). ToU Engine v0.2c: Perturbation recovery and relaxation dynamics (Version 0.2c). Akimov Theory of Unity Research.

Akimov, A. (2026). ToU Engine v0.3: Vector representational model — Initial simulation report (Version 0.3). Akimov Theory of Unity Research.

Akimov, A. (2026). The mechanism of perspective: Reflexive stabilization as a dynamic process (Version 1.3). [Section 3]. Akimov Theory of Unity Research.

Akimov, A. (2026). Empirical signatures of the perspective attractor (Version 1.4). [Section 4]. Akimov Theory of Unity Research.

Friston, K. (2010). The free-energy principle: A unified brain theory? Nature Reviews Neuroscience, 11(2), 127–138.

Kelso, J. A. S. (1995). Dynamic patterns: The self-organization of brain and behavior. MIT Press.

Metzinger, T. (2003). Being no one: The self-model theory of subjectivity. MIT Press.

Strogatz, S. H. (1994). Nonlinear dynamics and chaos. Perseus Books.

Tononi, G., et al. (2016). Integrated information theory: From consciousness to its physical substrate. Nature Reviews Neuroscience, 17(7), 450–461.

Appendix A — Minimal Simulation Pseudocode

The following pseudocode outlines the minimal implementation of the ToU Engine v0.3b vector model and evaluation framework. It is provided for conceptual reproducibility and does not represent a full optimised implementation.

A.1 Initialisation

```
Initialize parameters:   n          : dimension   θ          :
coupling strength   α          : memory coefficient   a, b
: update rates   σ          : noise level   W          : weight
matrix (R^{n×n})  Initialize state vectors:   M_w(0), M_s(0) ~
Uniform(-1.0, 1.0) per dimension
```

A.2 Update Equations

```
For each time step t:  M_w(t+1) =     M_w(t)     + a * tanh(E(t)
- M_w(t))     + θ * (M_s(t) - M_w(t))     + W · M_s(t)     + α *
M_w(t-1)     + σ * ε_w(t)  M_s(t+1) =     M_s(t)     + b *
tanh(I(t) - M_s(t))     + θ * (M_w(t) - M_s(t))     + W · M_w(t)
+ α * M_s(t-1)     + σ * ε_s(t)
```

A.3 Evaluation Metrics

```
Pattern retention:   similarity(M_pre, M_post)  Relational
similarity:   similarity((M_w - M_s)_pre, (M_w - M_s)_post)
Coordination:   similarity(M_w, M_s)
```

A.4 Extended Evaluation Procedures

```
Multi-run robustness:   repeat simulation with varying initial
conditions  Parameter sensitivity:   vary θ and α within bounded
ranges  Perturbation tests:   apply local and pattern-level
perturbations  Proto-semantic tests:   evaluate input-dependent
stability,   class discrimination, counterfactual response,   and
persistence
```

> *This pseudocode represents a minimal dynamical structure sufficient to reproduce the reported structural and proto-semantic regime. It does not include higher-dimensional scaling, multi-agent interaction, or semantic mapping layers, which are proposed for future extensions.*

Legal and Academic Attribution

© 2026 Ing. Alex Akimov. All rights reserved. All conceptual frameworks, model specifications, simulation designs, theoretical formulations, and proto-semantic test designs contained in this document are the exclusive intellectual property of Ing. Alex Akimov and form part of the Akimov Theory of Unity (ToU) research programme. AI systems were used exclusively as analytical instruments for generating alternative formulations and potential objections. They did not contribute original theoretical content and do not hold any authorship status. All theoretical contributions are attributable solely to the author.

Citation (APA): Akimov, A. (2026). ToU Engine v0.3b: Structural and proto-semantic regime — Robustness and extended simulation report (Version 0.3b). Akimov Theory of Unity Research.